

Solution Requirements

Date	03 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The system provides secure access to the application. Users can directly enter soil parameters to receive crop recommendations. Future versions will support farmer login and administrator authentication.	medium
2	Authorization levels	The application supports two user roles: Farmer/User – Predict suitable crops. Administrator – Manage datasets, ML models, and monitor predictions.	medium
3	External interfaces	Random Forest Machine Learning Model Scikit-Learn Library Flask Backend HTML, CSS and JavaScript Frontend Future Integration: Weather API and SHAP Explainability	High

4	Transactions processing	The application receives soil parameters, validates input, sends data to the ML model, predicts the crop, and returns the recommendation instantly.	High
5	Reporting	The system displays crop recommendation results, prediction status, model information, and future versions will generate downloadable prediction reports.	medium
6	Business rules, etc.	Crop prediction is generated only when all seven soil parameters are entered with valid values. The Random Forest model produces one recommendation based on trained agricultural data.	High
7	Compliance to laws or regulations	The application stores no personal information and follows standard software engineering and data privacy practices. Future cloud deployments will comply with applicable data protection regulations.	medium
8	Other	Future enhancements include: Weather Forecast Integration SHAP Explainability Confidence Score What-if Simulator Fertilizer Recommendation Multi-language Support	medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	The system should generate crop recommendations quickly using the trained Random Forest model.	<i>(e.g., Page loads in < 2 seconds)</i> Prediction response time < 2 seconds
2	Scalability	The application should support future deployment with multiple concurrent users and larger agricultural datasets.	Support 5,000+ users with expandable architecture
3	Security & Data Privacy	Validate all user inputs and protect application data against invalid requests and unauthorized access	Input validation and secure Flask backend implementation
4	Reliability & Availability	The application should consistently provide accurate crop predictions without system failure.	99% system availability
5	Usability & Accessibility	The interface should be simple, responsive, and easy for farmers and agricultural professionals to use.	Fully responsive interface with intuitive user experience
6	<i>Other</i>	The project should be maintainable, modular, portable, and easy to upgrade with new AI features.	Modular architecture supporting future feature integration